

Part I

SUMMARY of Related Rates

Summary

February 21, 2026

0.1 General steps for solving Related Rates problems.

- Introduce variables, identify the given rate and unknown rate.
- Draw a picture.
- Find equations.
- Differentiate with respect to t .
- Evaluate and solve.

Part II

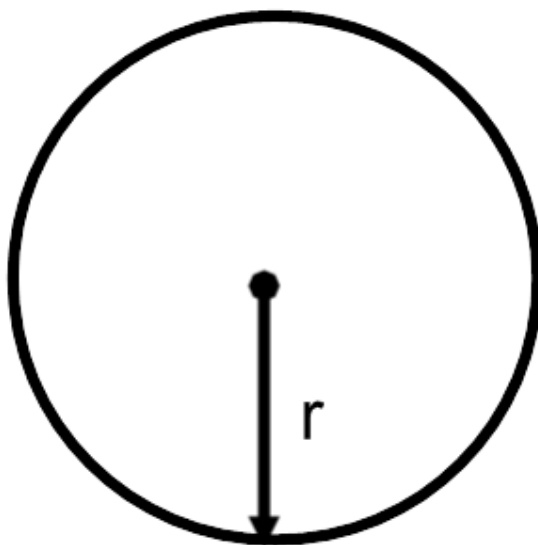
Recitation Questions

Problem 1

February 21, 2026

Problem 1

Problem 1 *The radius of a circle is increasing at a rate of 2 inches per minute.*



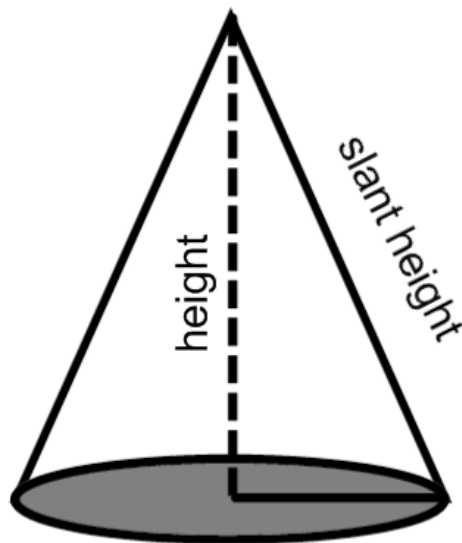
- (a) *At what rate is the circumference of the circle changing when the radius is 10 inches?*
- (b) *At what rate is the area of the circle changing when the radius is 12 inches?*

Problem 2

February 21, 2026

Problem 2

Problem 2 A right cone has a fixed slant height (see figure below) of 9 ft. The cone's height is shrinking at a rate of 0.5 ft/sec. At what rate is the **volume** of the cone changing when the height is 6 ft? Be sure to label the picture.

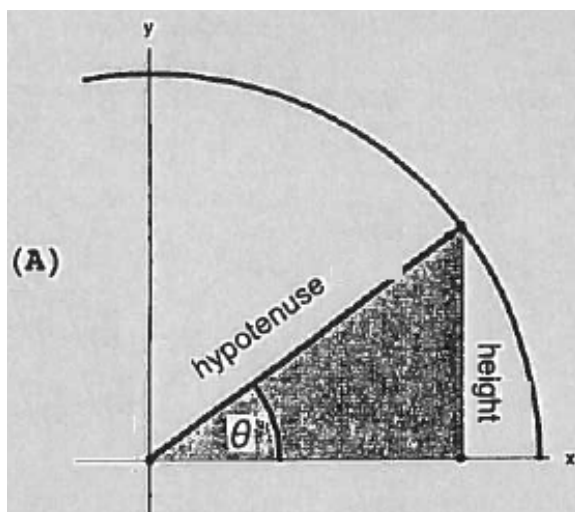


Problem 3

February 21, 2026

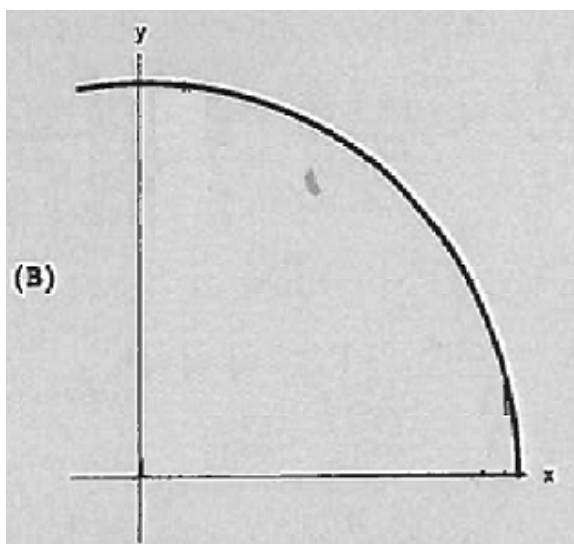
Problem 3

Problem 3 A part of a circle centered at the origin with radius $r = 7$ cm is given in the figure (A) below. A right triangle is formed in the first quadrant (see Figure (A)). One of its sides lies on the x -axis. Its hypotenuse runs from the origin to a point on the circle. The hypotenuse makes an angle θ with the x -axis. Assume that the angle θ changes at the rate $\frac{d\theta}{dt} = 0.2$ radians per second.



- (a) Label Figure A.
- (b) In Figure B, draw the triangle twice; once when θ is small and once more, when θ is close to $\frac{\pi}{2}$.

Problem 3



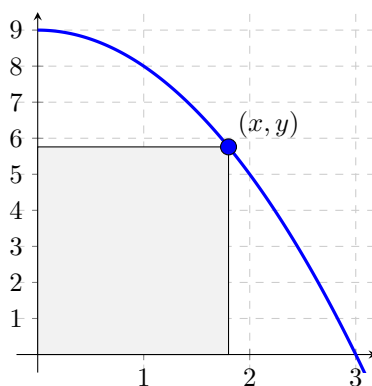
- (c) Find the rate of change of the height of the triangle when $\theta = \frac{\pi}{3}$
- (d) Find the rate of change of the area of the triangle when $\theta = \frac{\pi}{3}$

Problem 4

February 21, 2026

Problem 4

Problem 4 A rectangle in the first quadrant is constructed by taking a point (x, y) on the graph of the function $f(x) = 9 - x^2$, drawing a line segment vertically downward to the x -axis and a line segment horizontally leftward to the y -axis, as in the picture below. Denote the **length of the base** (along the x -axis) of the rectangle by x , the **height of the rectangle** (along the y -axis) by y (in meters), and the **PERIMETER** of the rectangle by P . When $x = 2$ m (and only at that moment), the height y is shrinking at a rate of $\frac{1}{5}$ m/s. Find the value of $\left[\frac{dP}{dt} \right]_{x=2m}$ by performing the steps below.



- Find a formula** for the height, y , of the rectangle as a function of x . (This is denoted as $y(x)$.)
- Find the value** of $\left[\frac{dx}{dt} \right]_{x=2m}$.
- Find a formula** for the perimeter, P , of the rectangle as a function of x . (This is denoted as $P(x)$.)
- Find the value** of $\left[\frac{dP}{dt} \right]_{x=2m}$.
- Write a sentence to explain** what the value found in (d) means about the rectangle. (Don't forget UNITS.)